

GSLC Session 4 - Machine Learning

Individual Task

Nama :

NIM :

Class :

Classifications Topics

1. What type of classification is the case below:

a) Apakah hari ini akan hujan?

x_1	x_2	x_3	x_4	Rain (y)
<i>value</i>	<i>value</i>	<i>value</i>	<i>value</i>	Yes/no

b) Ada buah apa saja didalam foto tersebut?



x_1	Grape (y_1)	Manggo (y_2)	Kiwi (y_3)	Apple (y_4)	Watermelon (y_5)	Banana (y_6)
<i>photos</i>	Yes/no	Yes/no	Yes/no	Yes/no	Yes/no	Yes/no

c) What kind of house suits my finances and how to pay for it?

x_1	x_2	x_3	x_4	House Type (y_1)	Payment method (y_2)
<i>value</i>	<i>value</i>	<i>value</i>	<i>value</i>	Type A/ Type B/ Type C/ Type D	Credit/Cash

d) What kind of sofa suits my living room?

x_1	x_2	x_3	x_4	Sofa (y)
<i>value</i>	<i>value</i>	<i>value</i>	<i>value</i>	Type A/ Type B/ Type C/ Type D

e) What should I wear today?

x_1	x_2	x_3	x_4	Top (y_1)	Pants (y_2)	Shoes (y_3)
<i>a value</i>	<i>a value</i>	<i>a value</i>	<i>a value</i>	t-shirts/ shirts/ suit/ coat	Jeans/ Cargo pants/ Slim-fit pants/ Tech pants	Combat boot/ Loafer/ Oxford shoe/ Sports shoes

- Make examples of each binary, multi-class, multi-label, and multi-output classification, such as question number one, accompanied by questions, tables, appropriate variable names for features(x) and labels (y), and possible answers for each target variable/labels (y).
- Guess the No Information Rate (NIR) value of this case:

Case1: I have a train and test set containing balanced data like this: Train set: Grape = 25 pictures Orange = 25 pictures Melon = 25 pictures Banana = 25 pictures Test set: Grape = 5 pictures Orange = 5 pictures Melon = 5 pictures Banana = 5 pictures	Case2: I have a train and test set containing unbalanced data like this: Train set: Grape = 70 pictures Orange = 10 pictures Melon = 10 pictures Banana = 10 pictures Test set: Grape = 14 pictures Orange = 2 pictures Melon = 2 pictures Banana = 2 pictures
Questions	
NIR =	NIR =
If I train a model with accuracy 60%. Is my model better than the NIR? (yes/no)	If I train a model with accuracy 60%. Is my model better than the NIR? (yes/no)

* Clue see ppt session 3 or 4 in slide 26

- Explain the differences between **Precision** and **Recall**. Why is it different? What is the purpose of using Precision and Recall?
- What is **F1 Score**? And explain when is it the right time to use the F1 score?



Regression Topics

6. Explain the differences between **Linear Regression** and **Polynomial Regression**.
7. Explain the differences between **underfitting** and **overfitting**.
8. coefficient of determination (R^2) value is always between 0 and 1. Explain what does it implied?
9. Can the coefficient of determination (R^2) be below zero? In practical sometime the (R^2) achieved below zero, then why does it happen and what does it mean?
10. We can use Pearson Correlation Coefficient (PCC) to see the correlation between two variables. The PCC value is always between -1 and 1. Explain what does it implied?

AOL Topics

11. Mention the names and NIMs of your AOL group members?
12. What type of Machine Learning Project will you perform as AOL project?
(Classification/Regression/Clustering/etc.)
13. Explain the problems you will raise in your AOL assignment?
14. Based on the Binus Sustainable Development Goals (SDG), What are the SDG topics that are suitable for your case?

